Chapter 5 — States and Phases of Matter

Board-style MCQs · PECTAA Chemistry 11 · Answer key at end

Practice MCQs

1. Ideal gas equation is:

(a) PV = nR/T **(b)** P/V = nRT **(c)** PT = nRV **(d)** PV = nRT

2. R in atm dm³ units:

(a) 1.987 **(b)** 6.02 **(c)** 8.314 **(d)** 0.0821

3. Gas pressure is due to:

(a) Chemical bonds **(b)** Wall collisions **(c)** Gravity only **(d)** Static atoms

4. Gases are highly:

(a) Incompressible **(b)** Fixed volume **(c)** Compressible **(d)** Fixed shape

5. Average kinetic energy depends on:

(a) Pressure only **(b)** Volume only **(c)** Absolute temperature **(d)** Molar mass only

6. London forces are also called:

(a) Ionic bonds **(b)** pd-pd forces **(c)** id-id forces **(d)** H bonds

7. Hydrogen bonding needs H bonded to:

(a) C or S **(b)** N, O or F **(c)** Cl only **(d)** Na or K

8. Water has high surface tension due to:

(a) London forces only **(b)** Metallic bonds **(c)** Ionic bonds **(d)** Hydrogen bonding

9. Ice is less dense than water due to:

(a) Close packing **(b)** Open lattice **(c)** Ionic lattice **(d)** No H bonds

10. Viscosity decreases with:

(a) Larger molecules **(b)** More H bonds **(c)** Cooling **(d)** Temperature rise

11. Evaporation causes:

(a) Heating **(b)** Cooling **(c)** No change **(d)** Freezing always

12. Vapour pressure increases with:

(a) Lower temperature **(b)** Cooling **(c)** Temperature **(d)** More solute always

13. Boiling occurs when vapour pressure equals:

(a) Density **(b)** External pressure **(c)** Atmospheric mass **(d)** Zero

14. Boiling point rises when external pressure:

(a) Increases **(b)** Is constant only **(c)** Is zero **(d)** Decreases

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15. Crystalline solids have:

(a) Softening range **(b)** No shape **(c)** Sharp melting point **(d)** Random arrangement

16. Amorphous solid example:

(a) NaCl **(b)** Ice **(c)** Diamond **(d)** Glass

17. Liquid crystals are used in:

(a) Only fuels **(b)** Only salts **(c)** Only metals **(d)** LCD screens

18. Stronger intermolecular forces give:

(a) Lower boiling point **(b)** No boiling **(c)** Lower viscosity always **(d)** Higher boiling point

19. At same T, lighter gas molecules move:

(a) Not at all **(b)** Same speed **(c)** Faster **(d)** Slower

20. Diffusion is fastest in:

(a) Crystals **(b)** Gases **(c)** Solids **(d)** Liquids

21. Surface tension is due to:

(a) Unbalanced surface forces **(b)** Ionic bonds **(c)** Wall collisions **(d)** Gravity only

22. Water expands on:

(a) Boiling only **(b)** Freezing **(c)** Melting **(d)** Condensing

23. HF has higher b.p. than HCl due to:

(a) No electrons **(b)** Ionic bonds **(c)** Hydrogen bonding **(d)** Lower mass

24. Viscosity is resistance to:

(a) Boiling **(b)** Flow **(c)** Melting **(d)** Diffusion only

25. A liquid with high vapour pressure is:

(a) Ionic **(b)** Volatile **(c)** Solid **(d)** Non-volatile

26. Molar volume at STP:

(a) 1 dm^3 **(b)** 22.4 dm^3 **(c)** 100 dm^3 **(d)** 24.4 dm^3

27. KMT assumes ideal gas particles have:

(a) Large volume **(b)** No motion **(c)** Negligible volume **(d)** Strong attraction

28. Permanent dipole forces occur in:

(a) Ions only **(b)** Polar molecules **(c)** Non-polar only **(d)** Metals only

29. Liquid crystals have properties between:

(a) Liquids and solids **(b)** Gases and plasma **(c)** Metals and salts **(d)** Acids and bases

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30. Boiling point of water at 1 atm:

(a) 273 °C **(b)** 0 °C **(c)** 50 °C **(d)** 100 °C

31. Gases are compressible because:

(a) High density **(b)** Large empty spaces **(c)** Strong bonds **(d)** Fixed positions

32. Diffusion of gases is due to:

(a) Random motion **(b)** Ionic bonds **(c)** Gravity only **(d)** Fixed lattice

33. Boyle's law relates:

(a) P and T **(b)** P and V **(c)** V and T **(d)** n and M

34. Charles' law relates:

(a) V and T **(b)** n and M **(c)** P and T **(d)** P and V

35. Avogadro's law relates:

(a) P and V **(b)** V and n **(c)** V and T **(d)** P and T

36. Ideal gas constant R in atm dm³ units:

(a) 0.0821 **(b)** 22.4 **(c)** 8.314 **(d)** 6.02

37. Kinetic theory says average KE depends on:

(a) Pressure only **(b)** Absolute temperature **(c)** Molar mass only **(d)** Volume only

38. London forces increase with:

(a) Less polarizability **(b)** Molecular size **(c)** Higher temperature only **(d)** Lower mass

39. Permanent dipole forces occur in:

(a) CO_2 **(b)** N_2 **(c)** HCl **(d)** CH_4

40. Water has high surface tension due to:

(a) Hydrogen bonding **(b)** No forces **(c)** Metallic bonding **(d)** Ionic bonding

41. Viscosity is:

(a) Surface pull **(b)** Vapour pressure **(c)** Resistance to flow **(d)** Boiling point

42. Evaporation occurs at:

(a) Only boiling point **(b)** Liquid surface **(c)** Solid lattice **(d)** Gas phase only

43. Vapour pressure increases with:

(a) Temperature **(b)** Lower temperature **(c)** Stronger forces **(d)** More solute always

44. Boiling point is when vapour pressure equals:

(a) Density **(b)** Zero **(c)** External pressure **(d)** Mass

45. Heat of fusion is for:

(a) Gas to liquid **(b)** Solid to gas **(c)** Solid to liquid **(d)** Liquid to gas

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46. Heat of vaporization is for:

(a) Solid to liquid **(b)** Liquid to gas **(c)** Liquid to solid **(d)** Gas to solid

47. Crystalline solids have:

(a) No melting point **(b)** Sharp melting point **(c)** Isotropy only **(d)** Random arrangement

48. Amorphous solids are:

(a) Always metallic **(b)** Always gas **(c)** Anisotropic only **(d)** Isotropic

49. Liquid crystals flow like:

(a) Plasma **(b)** Liquids **(c)** Gases only **(d)** Solids only

50. LCD uses:

(a) Metal vapour **(b)** Sand **(c)** Liquid crystals **(d)** Ionic melts

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Answer Key with Brief Reasons

1. **Answer: (d) PV = nRT** — Combined gas law.
2. **Answer: (d) 0.0821** — R = 0.0821 atm dm^3 K^{-1} mol^{-1}.
3. **Answer: (b) Wall collisions** — KMT explains pressure.
4. **Answer: (c) Compressible** — Large empty spaces.
5. **Answer: (c) Absolute temperature** — KE proportional to T.
6. **Answer: (c) id-id forces** — Instantaneous dipole-induced dipole.
7. **Answer: (b) N, O or F** — High electronegativity and lone pair.
8. **Answer: (d) Hydrogen bonding** — Strong surface attraction.
9. **Answer: (b) Open lattice** — H bonds create empty spaces.
10. **Answer: (d) Temperature rise** — Molecules move more freely.
11. **Answer: (b) Cooling** — High-energy molecules escape.
12. **Answer: (c) Temperature** — More molecules escape.
13. **Answer: (b) External pressure** — Bubbles can form throughout.
14. **Answer: (a) Increases** — Higher pressure needs higher VP.
15. **Answer: (c) Sharp melting point** — Ordered lattice.
16. **Answer: (d) Glass** — No long-range order.
17. **Answer: (d) LCD screens** — Optical response to field/temperature.
18. **Answer: (d) Higher boiling point** — More energy needed to separate.
19. **Answer: (c) Faster** — KE same, lower mass higher speed.
20. **Answer: (b) Gases** — Particles move freely.
21. **Answer: (a) Unbalanced surface forces** — Surface molecules pulled inward.
22. **Answer: (b) Freezing** — Open ice structure.
23. **Answer: (c) Hydrogen bonding** — Strong H bonds.
24. **Answer: (b) Flow** — Internal friction.
25. **Answer: (b) Volatile** — Escapes easily.
26. **Answer: (b) 22.4 dm^3** — One mole ideal gas.
27. **Answer: (c) Negligible volume** — Ideal assumptions.
28. **Answer: (b) Polar molecules** — pd-pd attraction.
29. **Answer: (a) Liquids and solids** — Intermediate order.
30. **Answer: (d) 100 °C** — Normal boiling point.
31. **Answer: (b) Large empty spaces** — Particles far apart.

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32. Answer: (a) Random motion — Kinetic theory.

33. Answer: (b) P and V — At constant T.

34. Answer: (a) V and T — At constant P.

35. Answer: (b) V and n — At constant P,T.

36. Answer: (a) 0.0821 — R=0.0821 atm dm^3 K^{-1} mol^{-1}.

37. Answer: (b) Absolute temperature — KE proportional to T.

38. Answer: (b) Molecular size — Larger electron cloud.

39. Answer: (c) HCl — Polar molecule.

40. Answer: (a) Hydrogen bonding — Strong cohesion.

41. Answer: (c) Resistance to flow — Internal friction.

42. Answer: (b) Liquid surface — High-energy surface molecules escape.

43. Answer: (a) Temperature — More molecules escape.

44. Answer: (c) External pressure — Boiling condition.

45. Answer: (c) Solid to liquid — Melting.

46. Answer: (b) Liquid to gas — Vaporization.

47. Answer: (b) Sharp melting point — Regular lattice.

48. Answer: (d) Isotropic — No long-range order.

49. Answer: (b) Liquids — Partial order.

50. Answer: (c) Liquid crystals — Optical switching.

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